

Week 6 Lab Assistant Preparation Guide

Important note first

I checked both uploaded repo ZIPs, and there is no week06/ folder in either repo.

The repo jumps from:

```
week05/  
week07/
```

So Week 6 materials are missing from the uploaded repo.

Because of that, I should not pretend there is an official Week 6 lab. Below is the practical prep guide you can use based on the gap between Week 5 and Week 7:

- Week 5 = LAN, Ethernet, static IP, ping, SCP/SSHFS
- Week 7 = IoT, Raspberry Pi temperature, HiveMQ Cloud, MQTT, Python publish/subscribe

So Week 6 is most likely either:

- a missing bridge week between networking and IoT, or
- a planned lab that was not added to the repo.

1. What Week 6 Is About

Since the official Week 6 folder is missing, the safest interpretation is:

Week 6 should prepare students for IoT networking

The likely focus should be:

- making sure Raspberry Pis are usable without constant monitor setup
- checking network access
- using SSH properly
- transferring files
- running Python scripts on the Pi
- understanding client/server communication before MQTT in Week 7

In simple terms:

Week 6 should make sure students can control the Raspberry Pi, connect it to a network, run Python programs, and understand how devices send messages before moving to cloud IoT.

2. Relevant Repo Folders and Files

There is no actual Week 6 folder.

Relevant surrounding folders:

```
week05/
```

Covers practical networking:

```
week05/README.md
week05/01_crimping_cable.md
week05/02_LAN.md
week05/03_test.md
week05/04_practical_app.md
week05/05_reflection.md
```

And:

```
week07/
```

Covers IoT and cloud:

```
week07/README.md
week07/01_temperature.md
week07/02_cloud_service.md
week07/03_python_pub_sub.md
week07/04_iot_service.md
```

Week 6 is missing between these two.

3. What Students Are Expected To Do

Because there is no Week 6 material, you should confirm with the module leader or staff what the official activity is.

But if you need to support the lab anyway, the best practical Week 6 activity would be:

Activity A - Confirm Raspberry Pi Access

Students should be able to:

- boot the Pi
- connect it to Wi-Fi or Ethernet
- find its IP address
- SSH into it from another computer

Useful commands on the Pi:

```
hostname -I
ip addr
```

From laptop/Mac/Linux:

```
ssh username@IP_ADDRESS
```

Example:

```
ssh pi@192.168.1.25
```

But remind students: newer Pis may not use the username pi.

Check username on the Pi:

```
whoami
```

Activity B - Basic Network Testing

Students should test internet/network connection:

```
ping -c 4 google.com
```

Or test local router:

```
ping -c 4 192.168.1.1
```

If DNS is the problem, this may fail:

```
ping google.com
```

But this may work:

```
ping 8.8.8.8
```

Simple explanation:

If pinging `8.8.8.8` works but `google.com` fails, the internet works but name lookup/DNS may be the issue.

Activity C - File Transfer With SCP

Students should transfer a Python file from laptop to Pi.

Example:

```
scp test.py username@PI_IP:/home/username/
```

Then SSH into the Pi and run it:

```
python3 test.py
```

This prepares them for Week 7 because they will need to create and run Python files on the Raspberry Pi.

Activity D - Run Simple Python Networking Code

A useful bridge before MQTT is a simple socket example.

Server on Pi A:

```
import socket

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind(("0.0.0.0", 5000))
server.listen(1)

print("Waiting for connection...")
conn, addr = server.accept()
print("Connected by", addr)

data = conn.recv(1024)
print("Received:", data.decode())

conn.sendall(b"Hello from server")
conn.close()
```

Client on Pi B or laptop:

```
import socket

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect(("SERVER_IP_ADDRESS", 5000))

client.sendall(b"Hello from client")
data = client.recv(1024)

print("Received:", data.decode())
client.close()
```

This helps students understand the idea of:

- one program waiting
- another program connecting
- IP address
- port number
- messages moving over a network

That is useful before MQTT in Week 7.

4. What You Should Personally Understand Before the Lab

SSH

SSH means Secure Shell.

Simple explanation:

SSH lets you control the Raspberry Pi from another computer using the terminal.

Students should know this structure:

```
ssh username@ip_address
```

Example:

```
ssh pi@192.168.1.25
```

Common issue:

```
Connection refused
```

Usually means SSH is not enabled or not running.

Check on the Pi:

```
sudo systemctl status ssh
```

Enable it:

```
sudo systemctl enable --now ssh
```

Or use:

```
sudo raspi-config
```

Then enable SSH from interface options.

IP Address

Simple explanation:

An IP address is the address of a device on the network. If your laptop wants to talk to the Pi, it needs the Pi's IP address.

Useful commands:

```
hostname -I  
ip addr
```

Ping

Simple explanation:

Ping is a quick test to check whether one device can reach another device.

Example:

```
ping -c 4 192.168.1.25
```

Important: without -c 4, Linux ping keeps running until students press:

```
Ctrl + C
```

SCP

SCP means Secure Copy.

Simple explanation:

SCP copies a file from one computer to another through SSH.

Structure:

```
scp local_file username@ip_address:/remote/folder/
```

Example:

```
scp script.py pi@192.168.1.25:/home/pi/
```

Python Scripts on Raspberry Pi

Students should know:

Create file:

```
nano script.py
```

Run file:

```
python3 script.py
```

Install package:

```
pip3 install package_name
```

But on some Raspberry Pi OS versions, they may get an "externally managed environment" error. If that happens, they may need:

```
python3 -m venv venv  
source venv/bin/activate  
pip install package_name
```

This will matter in Week 7 when they install:

```
pip3 install paho-mqtt
```

5. Common Student Mistakes and How To Fix Them

Mistake 1 - Using the Wrong Username

They may copy:

```
ssh pi@192.168.1.25
```

But their username may not be pi.

Fix:

On the Pi:

```
whoami
```

Then use:

```
ssh actual_username@192.168.1.25
```

Mistake 2 - Typing the Wrong IP Address

Students often mix up:

- their laptop IP
- the router IP
- the Raspberry Pi IP
- another group's Pi IP

Fix:

On the Pi:

```
hostname -I
```

Then write the IP down clearly.

Mistake 3 - SSH Is Not Enabled

Problem:

```
ssh: connect to host ... port 22: Connection refused
```

Fix:

```
sudo systemctl enable --now ssh
```

Check:

```
sudo systemctl status ssh
```

Mistake 4 - Ping Works But SSH Does Not

This means the network connection is probably fine, but SSH is not running.

Fix SSH, not the network.

```
sudo systemctl status ssh
```

```
sudo systemctl enable --now ssh
```

Mistake 5 - SCP Destination Path Is Wrong

Problem:

```
scp file.py pi@192.168.1.25:/home/pi/
```

fails because /home/pi/ does not exist.

Fix:

Use the correct username/path:

```
scp file.py username@192.168.1.25:/home/username/
```

Or just send to the home folder:

```
scp file.py username@192.168.1.25:~
```

Mistake 6 - Python File Saved With Wrong Name

Students may save:

```
script.py.txt
```

or run the wrong file.

Fix:

Check files:

```
ls
```

Run:

```
python3 script.py
```

Mistake 7 - Running Client Before Server

For socket programs, the server must run first.

Simple explanation:

The server is waiting for a call. The client makes the call. If nobody is waiting, the call fails.

Correct order:

1. Start server
2. Start client

3. Check messages

Mistake 8 - Wrong Port Number

If server uses port 5000, client must also use 5000.

Bad:

```
server.bind(("0.0.0.0", 5000))
client.connect(("192.168.1.25", 6000))
```

Good:

```
server.bind(("0.0.0.0", 5000))
client.connect(("192.168.1.25", 5000))
```

Mistake 9 - Firewall or Network Blocking

If students use university Wi-Fi, devices may not be allowed to talk to each other.

Fix options:

- use direct Ethernet connection
- use a local router/hotspot
- use the same lab network if peer-to-peer traffic is allowed
- test with `ping` before debugging Python

6. Commands and Concepts You Should Be Ready To Explain

Network Commands

Show IP addresses:

```
hostname -I
```

Detailed network info:

```
ip addr
```

Show Ethernet only:

```
ip addr show eth0
```

Show Wi-Fi only:

```
ip addr show wlan0
```

Ping another device:

```
ping -c 4 IP_ADDRESS
```

SSH Commands

Connect:

```
ssh username@IP_ADDRESS
```

Check SSH service:

```
sudo systemctl status ssh
```

Enable SSH:

```
sudo systemctl enable --now ssh
```

File Transfer Commands

Copy file to Pi:

```
scp file.py username@IP_ADDRESS:~
```

Copy file from Pi to laptop:

```
scp username@IP_ADDRESS:~/file.py .
```

Python Commands

Check Python version:

```
python3 --version
```

Run script:

```
python3 script.py
```

Install package:

```
pip3 install package_name
```

Create virtual environment if needed:

```
python3 -m venv venv  
source venv/bin/activate  
pip install package_name
```

7. Physical or Software Setup To Check Before the Session

Physical Setup

Check:

- Raspberry Pis are available
- power supplies are available
- microSD cards are inserted
- monitors/keyboards/mice are available if SSH fails
- Ethernet cables are available
- spare known-working Ethernet cables are available
- Wi-Fi credentials are known if students need Wi-Fi
- students know which Pi belongs to which group

Software Setup

Check:

```
python3 --version
```

Check SSH:

```
sudo systemctl status ssh
```

Check network:

```
hostname -I
```

Check package installation:

```
pip3 --version
```

If Week 7 is next, you can also check whether this works:

```
pip3 install paho-mqtt
```

or in a virtual environment:

```
python3 -m venv venv
source venv/bin/activate
pip install paho-mqtt
```

8. Simple Explanations You Can Give To Beginners

What is SSH?

SSH lets you control the Raspberry Pi from another computer using the terminal.

What is an IP address?

It is the network address of the device. Your laptop needs the Pi's IP address to connect to it.

What is ping?

Ping checks if two devices can reach each other.

What is SCP?

SCP copies files between computers using SSH.

What is a port?

A port is like a door number inside a device. The IP address finds the device, and the port finds the program inside that device.

What is a server?

A server waits for another device to connect.

What is a client?

A client starts the connection to the server.

Why learn sockets before MQTT?

Sockets show the basic idea of devices sending messages. MQTT is a more organised system built for sending messages between IoT devices.

9. Short Checklist For You Before the Lab

Repo/Content

- Confirm with staff that Week 6 is missing from the repo
- Ask whether Week 6 has a separate document, slide deck, or verbal plan
- Ask whether Week 6 should be a catch-up/bridge lab
- Confirm whether students should submit anything for Week 6

Raspberry Pi Setup

- Pi boots correctly
- Pi has network access
- SSH is enabled
- Username/password are known
- Python 3 works
- Students can create/run `.py`` files
- File transfer with `scp`` works

Commands To Keep Ready

```
hostname -I
ip addr
ping -c 4 IP_ADDRESS
ssh username@IP_ADDRESS
scp file.py username@IP_ADDRESS:~
sudo systemctl status ssh
sudo systemctl enable --now ssh
python3 --version
python3 script.py
```

10. Missing or Incomplete Parts in the Repo

The main issue is:

```
week06/ is missing
```

The README also does not mention Week 6. It lists Week 5 and then Week 7.

This means:

- there are no Week 6 instructions
- there are no Week 6 student evidence files
- there is no Week 6 README
- there is no official Week 6 objective in the repo
- there is no submission template for Week 6

You should clarify this with staff before the lab.

Suggested message to staff:

```
Hi, I checked the lab journal repo and noticed there is no week06 folder. The
repo goes from week05 to week07. Could you please confirm whether Week 6 has
separate materials, or whether it is intended to be a catch-up/bridge lab before
the IoT and MQTT work in Week 7?
```

11. Best Practical Plan If Staff Say "Use Week 6 As Catch-Up"

If Week 6 is a catch-up or preparation week, I would run it like this:

First 20 minutes

Check every group can access their Pi:

```
hostname -I
ssh username@IP_ADDRESS
```

Next 20 minutes

Check file transfer:

```
scp test.py username@IP_ADDRESS:~  
python3 test.py
```

Next 30 minutes

Run a simple socket server/client.

Last 20 minutes

Prepare for Week 7:

```
python3 --version  
pip3 --version
```

Explain:

- Raspberry Pi temperature reading
- Python scripts
- cloud MQTT broker
- publisher and subscriber idea

12. Main Thing To Remember

Do not present Week 6 as if it exists in the repo.

The honest answer is:

Week 6 is missing from the uploaded repo. The best lab assistant preparation is to treat it as a bridge/catch-up session between Week 5 networking and Week 7 IoT, unless staff provide separate Week 6 materials.